CrewForge — race-proof team formation at scale

Hack4Hack Crew Hiring Challenge 2026 · PS1 · Database-enforced membership for high-concurrency hackathons

Problem. When many participants compete for the final team seat, read-then-write APIs can overfill teams, duplicate membership or corrupt state after retries. Client-side checks cannot protect shared state across app instances.

Product workflow

Discover teams by skills and open seats → **Create** a team with capacity and skill needs → **Request** a seat through an idempotent pending request → **Approve** or decline by the current leader → **Maintain** membership, invites and leadership succession. Pending requests do not reserve seats.

Architecture and correctness

Next.js API routes enforce authentication, validation, origin checks and rate limits. MongoDB stores users, teams, memberships, join requests, invites and audit events. A leader approval verifies authority and request state, inserts membership, conditionally increments `memberCount` only below `maxSize`, marks the request accepted and appends an audit event in one transaction. Unique compound indexes enforce one membership per participant per hackathon. Any failed guard aborts the transaction.

Concurrency proof

A MongoDB replica-set integration test submits **50 pending requests** and runs **50 concurrent leader approvals** for one remaining seat. Result: **1 accepted, 49 rejected**; 2 memberships including the leader; `memberCount = 2`; exactly one acceptance audit event. The test invokes the same transaction helpers used by the API routes.

Verified scale and latest CI benchmark

CI verifies **10,000 synthetic users, 2,000 teams, 6,000 memberships, 4,000 unassigned participants and 4,000 open seats**, plus two judge-only accounts. These are fixtures, not real registrations.

71 / 71	98.43	10.23 ms	0%	6,000/6,000
automated tests	requests / sec	p95 latency	HTTP failures	board checks

100 virtual users · 30 seconds · 3,000 requests · 0% failures · 6,000 board checks. GitHub-hosted runner + disposable MongoDB 7 replica set; synthetic regression benchmark, not production capacity. CI also passed typecheck, production build, dependency audit, five smoke checks and transaction-kill rollback.

Security and lifecycle

HttpOnly sessions · server-side leader authorization · same-origin mutation checks · persistent rate limits · hashed expiring invite/auth tokens · transactional audit events · bounded SSE with reconnect fallback.

Trade-offs and release boundary

Transactions and unique indexes keep invariants at the source of truth but require a MongoDB replica set. SSE is simpler than WebSockets for one-way invalidation but change streams consume connections. The public Vercel URL is an interactive browser-only preview; the full-stack app runs in local staging and is not publicly deployed. Production launch still requires MongoDB Atlas, verified Resend sender configuration and staging verification.

Evidence: CI 42057405278 · CodeQL 42057405246 · Repository · Public preview